

6. You have been given three jars containing the compounds **potassium bromide**, **barium chloride** and **calcium iodide**.



Unfortunately the labels have been removed from the jars and, because they are all white compounds, they cannot be identified by their appearance.

Describe **two** tests that you should carry out **on all three** of the compounds, so that each compound can be identified.

State how you would carry out both tests and give the expected observations for each compound. [6 QER]

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6. Describe how respiration and photosynthesis keep the carbon dioxide and oxygen content of the atmosphere approximately constant. Discuss how human activity is threatening this balance.
[6 QER]

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6. Several areas of the UK add fluoride to drinking water.

State the benefit of fluoridation and discuss the reasons why some people are opposed to it.
[6 QER]

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6. (a) A technician carried out a flame test and a silver nitrate test on a solution of sodium iodide.

Draw **one** line from **each** test to the correct observation.

[2]

Test	Observation
flame test	red flame
	yellow flame
	lilac flame
silver nitrate test	white precipitate
	cream precipitate
	yellow precipitate

- (b) (i) When silver nitrate and sodium iodide react, sodium nitrate and a precipitate of silver iodide are formed.

Write the formulae of sodium nitrate and silver iodide to complete the equation. [2]



- (ii) Suggest a method you could use to remove the precipitate from the reaction mixture.

[1]

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6. Fluoride ions (mostly from sodium fluoride) are found naturally in the drinking water in some regions of England. The concentration of these fluoride ions is quite low with values being less than 0.5 mg/dm^3 .

However, water companies in some areas add fluoride ions to drinking water. Areas where this happens include Yorkshire, the North East, the East Midlands and the West Midlands. The mean concentration of fluoride ions added is around 1 mg/dm^3 .

It is known that a high concentration of fluoride ions in drinking water can be poisonous and can cause a yellowing of teeth – fluorosis. Fluoridation of drinking water is a controversial issue. Some people are opposed to it because it removes the individual’s choice to not take fluoride.

The table shows the mean number of decayed, missing and filled teeth (DMFT) in five-year-old children in some areas of England in 2018.

Area	Mean DMFT
South East	0.6
West Midlands	0.7
North West	1.3
London	1.0
South West	0.7
Yorkshire	1.0
North East	1.0
East Midlands	0.9
East England	0.7



(a) Which **two** of the following statements are correct according to the information given?
Tick (✓) **two** boxes. [2]

- London, Yorkshire and the North West all have a mean DMFT of 1.0 ☐
- The South East has the lowest mean DMFT ☐
- The East Midlands has the highest mean DMFT ☐
- East England, the South West and the South East have a mean DMFT of 0.7 ☐
- The East Midlands has a lower mean DMFT than Yorkshire ☐
- London, the North West and the South West all have a mean DMFT of 0.9 ☐

(b) (i) Which of the following statements supports the argument for fluoridation of drinking water? Tick (✓) **one** box. [1]

- All areas with fluoridated drinking water have lower mean DMFT than areas which do not fluoridate drinking water ☐
- All areas with fluoridated drinking water have the same mean DMFT as non-fluoridated areas ☐
- All areas with fluoridated drinking water have lower mean DMFT than the North West ☐

(ii) Which of the following statements could be used to oppose the argument for fluoridation of drinking water? Tick (✓) **one** box. [1]

- It is not possible to decide whether fluoridation of drinking water lowers mean DMFT ☐
- All areas with fluoridated drinking water have higher mean DMFT than areas which do not fluoridate drinking water ☐
- All areas with fluoridated drinking water have a higher mean DMFT than the South West ☐



(c) Suggest a possible reason why the South East has a low mean DMFT in five-year-old children even though no fluoride is added to drinking water in this area. [1]

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